TVERSE : A DIGITAL TUTORING ECOSYSTEM TO ENHANCE LEARNING ACCESS AND EMPOWER ASPIRING TUTORS

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ABSTRACT: This study proposed a new digital business platform in empowering students and aspiring tutors called Tverse. In this competitive era of digital transformation, there are still many graduates who remain unemployed, B40 students who struggle financially to support their studies, and learners with low academic performance striving to achieve better results. This paper offers the conceptual ideas of Tverse, multi-sided digital platforms in solving the problems mentioned by leveraging United Nations Sustainable Development Goals (SDGs): no poverty, quality education (SDG 1 and SDG 4). Tverse is conceptualised as a marketplace that provides specific pain relievers, such as affordable, on-demand tutoring for learners and gain creators, such as flexible income, a platform to build a reputation for tutors to address the essential needs of its customer segments. This paper discussed the design thinking approach while building conceptual business models using modelling tools: Business Model Canvas (BMC), Value Proposition Canvas (VPC), Environmental Map (EM), and Strategic Canvas (SV). This approach comprises conducting a comprehensive literature review, benchmarking against relevant best practices, prototyping, and carrying out surveys and interviews to identify the needs, challenges, and key problems.

Keywords: B40, Sustainable Development Goals, Multi-sided platform, Business Model Canvas, Value Proposition Canvas, Environmental Map, Strategic Canvas.

1. INTRODUCTION

In today's digital economy, graduate unemployment remains a critical concern despite the abundance of online opportunities. Many talented individuals lack the entrepreneurial and technological tools to transform their knowledge into viable income sources. *TVerse* aims to address this gap by leveraging technopreneurship principles to connect, empower, and sustain both students and aspiring tutors in a dynamic digital learning ecosystem. According to Subbiah (2023), " Due to the technological advancements that have occurred in the past couple of years, it has become easier for people to access the gig economy"(p.1). The gig economy is described as a labor market that relies on freelancing and short-term contracts rather than permanent, full-time positions. This model is characterized by the prevalence of temporary work assignments and flexible jobs (Sachdeva, 2024). Rashid et al. (2023) argue that the gig economy provides earning opportunities for Malaysia's B40 group, supporting their socio-economic sustainability. Key factors in this include workload, risk, prospects, and circumstances; flexibility was not found to be a significant factor.

Hence, Tverse is introduced as an innovative platform for tutoring, which falls under the gig economy category. The platform provides opportunities for individuals to engage in freelancing, temporary work, or short-term tutoring jobs, allowing tutors and students to connect flexibly according to their availability and needs. By aligning with the gig economy model, Tverse promotes autonomy, skill-based employment, and inclusive access to educational support. For students, Tverse addresses their need to find reliable tutors, learn at their own pace, and access affordable and flexible learning options through a trustworthy digital environment. Meanwhile, for tutors, the platform enables them to connect with students who require their subject expertise, manage their tutoring schedules efficiently, and receive fair and timely payments, empowering them to work independently and sustain flexible income sources. Kufeoglu (2022) mentioned, “Education is a component of sustainable development with its strong effects at global, regional, and local levels”. This initiative aligns with Sustainable Development Goal 4 (Quality Education), which emphasizes inclusive and equitable access to lifelong learning opportunities. By promoting digital learning and skill development through the gig economy, Tverse contributes to building a more educated, empowered, and sustainable society.

Besides, Tverse incorporates several key features designed to address the pain points of both students and tutors within the tutoring ecosystem. For students, the platform offers verified tutor profiles, ensuring that every tutor undergoes a thorough review process before being listed. This feature enhances trust and reliability, helping students feel confident in selecting qualified tutors. Additionally, a tutor rating and feedback system enables learners to share their experiences and evaluate teaching quality, creating a transparent and accountable learning environment. On the other hand, for tutors, Tverse integrates a smart matching algorithm that connects them efficiently with students seeking their specific subject expertise, reducing the time spent searching for suitable learners. Furthermore, a profile builder and rating system allows tutors to showcase their qualifications, teaching style, and student feedback to help them strengthen their reputation, attract more learners, and build sustainable careers within the gig economy.By providing a platform where unemployed graduates can become tutors, Tverse contributes to Sustainable Development Goal 1 (No Poverty) by creating new income-generating opportunities. This initiative not only alleviates financial challenges but also empowers graduates to transform their academic skills into meaningful and sustainable livelihoods.

Many existing tutoring platforms, both locally and globally, face weaknesses such as inconsistent tutor quality, limited personalization, high service fees, and a lack of inclusivity for lower-income users. Some rely heavily on manual matching or high commissions that disadvantage tutors, while others offer limited engagement tools, making learning less interactive and transparent. These issues often exclude students from B40 communities and unemployed graduates who lack access to affordable or fair digital learning opportunities. TVerse addresses these gaps by creating a more accessible, data-driven, and inclusive ecosystem. Through verified tutor profiles, smart matching algorithms, and personalized dashboards, TVerse ensures reliable connections between students and tutors. Its affordable, transparent, and flexible model empowers B40 students to learn without financial strain while enabling unemployed graduates to earn sustainably and build professional credibility in a supportive digital environment.TVerse's gain creators enhance the experience for both students and tutors by promoting convenience, trust, and growth. For students, the platform recommends the best tutors based on their learning needs, allows flexible scheduling to reschedule or book sessions easily, and ensures safe transactions with quick refunds when issues arise. Students can also track their learning progress and feedback history, fostering continuous improvement. For tutors, TVerse builds credibility by listing only qualified tutors, provides insights on ratings, earnings, and engagement. This will support professional growth and guarantee fast, fair payment processing, creating a trusted and sustainable digital tutoring ecosystem.

2. PROBLEM STATEMENTS/OBJECTIVES

The Malaysian education landscape shows that while digital learning is expanding, B40 students are still left behind due to limited access to devices, internet connectivity, and affordable tutoring options because of financial issues. Based on the Department of Statistics Malaysia (2017), the monthly income of B40 families does not exceed RM4,850. This shows that most of the B40 students were facing barriers to access quality education and learning opportunities compared to those students from higher-income households who can afford tutoring. Nowadays, tutoring services are like a privilege for those who are wealthy enough to afford it, which will further widen the social inequalities in Malaysia. Chang and Weerasena (2017) mentioned that parents who have a higher level of educational background, which implies a higher level of income, tend to spend on tuition sessions for their children. Moreover, digital and infrastructural inequalities in education become a barrier for B40 students at both secondary and higher education levels to gain access to flexible learning. Based on the research conducted by Devisakti et al. (2022), a total of 511 respondents had answering the survey regarding the digital learning tools that they are using, majority of the respondents (80%) are using laptops, while others said they are using smartphones (14.2%), desktop computers (2.9%), and tablets (2.9%), highlighting unequal access to suitable devices which may limit their ability to fully engage efficiently in online learning activities and doing assignments. From these findings, a considerable portion still depend on less efficient devices, particularly those from lower-income households that struggle to afford proper learning devices. The B40 students also face socioeconomic challenges that can reduce their learning motivation and limit access to qualified and verified educators on the tutoring platform. Masalin and Saibeh (2025) quoted that "while families with higher socioeconomic status (SES) can afford private tutoring, those with lower SES face financial barriers" (p. 1).

This paper also focused on educators, especially underemployed graduates, who are currently facing challenges in securing stable employment and financial issues. Fresh graduates may face limited career pathways and professional exposure, which may force them to accept jobs that require a lower level of qualification compared to their actual educational level, rather than waiting for jobs that are equal to their qualifications, just to gain income (Shahidan et al., 2019). Next, underemployed graduates and part-time workers lack sustainable side income options. Based on Siew and Sze (2025), this problem is due to the graduates who frequently work beneath their qualification level, which may reduce their wages and lead to financial issues. Furthermore, the graduates among the tutors might not have a reputation-building platform or verified ratings for better employability. The Tverse platform can help address this problem, which aligns with the 13th Malaysia Plan (13MP), which is to emphasize the need to better align labour supply and demand, while preventing the underemployment issue among the graduates.

The main objective of this paper is to develop a conceptual business model for a digital tutoring platform that addresses the key challenges faced by both B40 students and unemployed graduates. Specifically, this study aims:

a. To develop a conceptual business model for a tutoring platform that supports B40 students and unemployed/underemployed graduates.
b. To help B40 students easily find and book qualified tutors while ensuring they gain meaningful knowledge and learning outcomes from tutoring sessions.
c. To encourage unemployed/underemployed graduates who possess teaching credentials or academic expertise to become part-time tutors and share their knowledge through the platform.
d. To create opportunities for unemployed and underemployed graduates to participate in digital entrepreneurship through tutoring while enhancing their income and employability.

3. METHODOLOGY

This research uses design thinking to tackle urgent problems such as poor access to education, unemployment, and uneven tech availability, particularly in Malaysia's lower-income B40 group. Since design thinking puts people first while pushing practical solutions, it helps develop innovative ideas that respond directly to users' challenges (Che Noh & Abdul Karim, 2022). By applying this approach, Tverse seeks to design an inclusive and accessible platform that bridges the gap between learners and tutors through technology. To shape progress clearly, multiple techniques came into play, such as Environmental Mapping, Business Model Canvas(BMC), Value Proposition Canvas(VPC), Literature Review, Strategic Canvas, Interviews, and Online Surveys. Together, these methods help spot learner demands, support the validation of business assumptions, and also ensure Tverse is able to deliver something meaningful within Malaysia's current learning landscape (Devisakti et al., 2023).

A. Business Model Canvas (BMC)

The Business Model Canvas (BMC) works like a roadmap for companies, helping them see and adjust how they build, share, and earn value within a single framework. Instead of just outlining an idea, it pushes teams to dig into customer relationships, key activities, and long-term money flows, enabling teams to make data-driven adjustments (Salwin et al., 2022). When Tverse uses this model, it lines up educational goals with operational and financial feasibility, so both tutors and learners get what they need. This approach also helps the project identify value creation opportunities and strategic improvements before full-scale implementation.

B. Value Proposition Canvas (VPC)

The Value Proposition Canvas (VPC) works like a direct instruction to match what companies offer with what customers actually need by looking closely at jobs-to-be-done, pains, and gains on one side, and determining how products or services act as pain relievers and gain creators on the other (Kyhnau, J., & Nielsen, C., 2015). When startups pay attention to this connection, it helps reduce market risk and improve product-market alignment. In the context of Tverse, using this method shows exactly where it steps to fix hassles faced by tutors and learners; it functions as a key conjunction between user empathy and viable service design, ensuring that Tverse's value proposition resonates deeply with both learners and tutors.

C. *Literature Review*

The Literature Review in this study was conducted to explore existing research related to educational technology, Design Thinking, and digital inequality. Checking earlier findings helps spot core theories, successful implementation strategies, and missing pieces this research wants to fix (Snyder, 2019). Through an extensive analysis of scholarly articles, reports, and case studies, this review supports the formulation of a relevant evidence-based framework for Tverse's design development. Plus, it makes sure the new approach fits today's learning upgrades and digital challenges highlighted in previous research.

D. Interviews / Surveys

The team used interviews along with surveys to gather detailed and numerical data. Talking one-on-one helped uncover personal views, reasons behind actions, and contextual barriers that may not be captured through quantitative means (Creswell & Poth, 2018). On the flip side, written questionnaires reached more folks quickly, showing trends while backing up what came out in conversations (Bryman, 2016). Using both ways at once strengthened results by cross-checking facts - making outcomes about user needs and Tverse design choices more solid.

4. LITERATURE REVIEW

4.1. The Digital Megatrend: 4IR, COVID-19 Impact, and National Digital Agendas

The operating environment for Tverse is fundamentally shaped by the confluence of the Fourth Industrial Revolution (4IR) and the lasting effects of the COVID-19 pandemic, both of which have accelerated digital transformation in education and the labor market (Nguyen & Nguyen, 2024; APEC, 2020).

Acceleration of Digital Transformation and 4IR: The 4IR, characterized by advances in AI, IoT, and intelligent systems, has rapidly permeated various sectors, creating a demand for new skills and the inevitable automation of routine tasks in both manufacturing and white-collar environments (APEC, 2020). This megatrend has driven an accelerated adoption of digital learning tools, including intelligent tutoring systems and virtual reality, even in developing countries that previously lagged behind (ResearchGate, 2023). Technopreneurship, which integrates technological expertise with a business vision, is seen as the primary mechanism for fostering innovation and competitive business models in this new era (Basly & Hammouda, 2020).

Impact of COVID-19 on Education and Employment: The pandemic forced a rapid and unprecedented shift to remote learning, highlighting the potential of digital platforms to minimize educational disruption (Ignacio et al., 2022). While this shift created opportunities for personalized learning and enhanced accessibility, it also exposed significant digital divides in technology access and infrastructure, especially in developing regions (Frontiers, 2025). Economically, COVID-19 caused staggering job losses and further complicated post-pandemic recovery by encouraging the accelerated deployment of 4IR automation technologies, intensifying job precarity and income uncertainty (APEC, 2020).

National Digital Agendas (e.g., MyDigital, NEP 2030): National initiatives are designed to harness the 4IR and digital economy for economic growth. These plans typically focus on building a resilient digital infrastructure and upskilling the workforce. Tverse aligns with the spirit of these agendas by promoting digital entrepreneurship and enhancing the digital competency of both students and tutors (Jamaludin et al., 2024).

4.2. Conceptual Framework: Technopreneurship and Multi-Sided Platforms

Tverse, as a new digital business platform, is an example of technopreneurship that employs a multi-sided platform model.

Technopreneurship and Digital Innovation: Technopreneurship is a critical element of the innovation system, restructuring business patterns and driving new forms of value creation (MDPI, 2020). The success factors for technopreneurial ventures include risk-taking, creativity, and the ability to translate technological innovation into commercially viable outcomes (Mihajlović et al., 2022). Tverse's conceptual model directly addresses the need for such innovation by leveraging technology to solve socio-economic problems related to education and income.

Multi-Sided Platform Business Models: Digital tutoring platforms operate as multi-sided platforms, mediating value exchange between two or more distinct but interdependent customer segments. In this case, students (learners) and tutors (service providers) (Cusumano, Yoffie, & Gawer, 2020). Platform research highlights the importance of business model innovation, specifically changes in target customers' roles and the revenue model to adapt to the commercial activities of its users (Micheli, Berchicci, & Jansen, 2020). The use of the Business Model Canvas (BMC) is widely recognized as an effective and simplified visual tool for designing, communicating, and testing the core logic and feasibility of such innovative start-up models (Türko, 2016).

4.3. Educational Disparity and The B40 in Malaysia

Tverse’s focus on the B40 (Bottom 40%) income group is critical, as this community faces significant barriers to accessing quality education and supplemental learning resources, contributing to pronounced educational disparity (Ministry of Education Malaysia, 2022).

Academic Achievement Gap and Affordability: Studies consistently show a correlation between socio-economic status and academic performance, resulting in a persistent achievement gap (Ismail et al., 2020). While private tutoring is a common resource to bridge this gap, its high cost is often prohibitive for B40 households, forcing students to rely solely on public schooling, which may not always be sufficient to meet individual learning needs (World Bank, 2020). Tverse directly addresses this by aiming to provide an affordable tutoring solution, linking to the goal of SDG 4: Quality Education.

The Digital Divide and Infrastructure: The shift to digital learning, accelerated by COVID-19, exposed the digital divide, where B40 students often lack reliable internet access, suitable devices, and a conducive home environment for online learning (UNICEF Malaysia, 2021). Tverse must design its platform and delivery model to be accessible and cost-effective, mitigating the digital divide rather than exacerbating it. This includes considering low-bandwidth options or mobile-friendly interfaces.

Tackling No Poverty (SDG 1): By simultaneously addressing the B40's need for affordable education and the unemployed graduates' need for income, Tverse's conceptual model aligns perfectly with both SDG 1 (No Poverty) and SDG 4. The platform acts as a socioeconomic bridge, moving beyond a purely commercial model to one with dual social impact (Jamaludin et al., 2024).

4.4. The Gig Economy and Graduate Unemployment

The rise of the Gig Economy, driven by digital platforms, represents a significant shift in labor market structures, offering flexible work arrangements and new income streams that bypass traditional employment rigidities (Lund et al., 2020). For platforms like Tverse, this model is key to addressing the challenge of graduate unemployment and underemployment.

Digital Platforms as Job Multipliers: Digital platforms have become crucial intermediaries, facilitating peer-to-peer services and creating micro-entrepreneurship opportunities (Abdullah & Lim, 2021). This model is particularly attractive to unemployed or underemployed graduates who possess valuable skills but struggle to find full-time conventional jobs that match their qualifications (Pew Research Center, 2021). By operating on a platform, they can immediately monetize their academic expertise as part-time tutors, enhancing their income and improving their employability through practical experience (Ameen, 2020).

Technopreneurship in the Gig Economy: Engaging in platform-based tutoring requires graduates to adopt a technopreneurial mindset, managing their time, marketing their services, and leveraging the platform's digital tools. This participation helps them acquire soft skills and digital competencies, which are highly valued in the 4IR job market (Hooi, 2020). Tverse, by facilitating this, serves as both an income-generating tool and an incubator for digital entrepreneurship among the skilled workforce.

4.5 Budget 2026

Malaysia's Budget 2026 presents a highly favorable policy environment for a platform like Tverse, particularly due to its strong emphasis on expanding access to education and strengthening digital infrastructure. The budget allocates RM 66.2 billion to the Ministry of Education, the largest share among ministries, signaling a deepening commitment to educational reform and access.

Meanwhile, RM 7.9 billion of the budget is earmarked for Technical and Vocational Education and Training (TVET), with a specific focus on integrating artificial intelligence (AI) and digital skills among learners. This governmental push toward digital education aligns closely with Tverse's mission of providing accessible, technology-driven tutoring to underserved segments, including underprivileged students.

Furthermore, Budget 2026 also includes targeted support for low-income students: about RM 2 billion is allocated to aid families from B40 households, students with special needs, and underprivileged schoolchildren to ensure more equitable access to educational resources. The emphasis on digital inclusion is also reflected in infrastructure investments, as companies like Telekom Malaysia have welcomed the government's plans to expand digital connectivity, including submarine cables and AI centres. These policy directions provide a strong foundation for Tverse to scale as a socially inclusive edtech platform one that not only supports learners from marginalized backgrounds, but also aligns with national priorities around digital literacy, AI-powered learning, and lifelong skills development.

4.6 Benchmarking of business models

1. Superprof

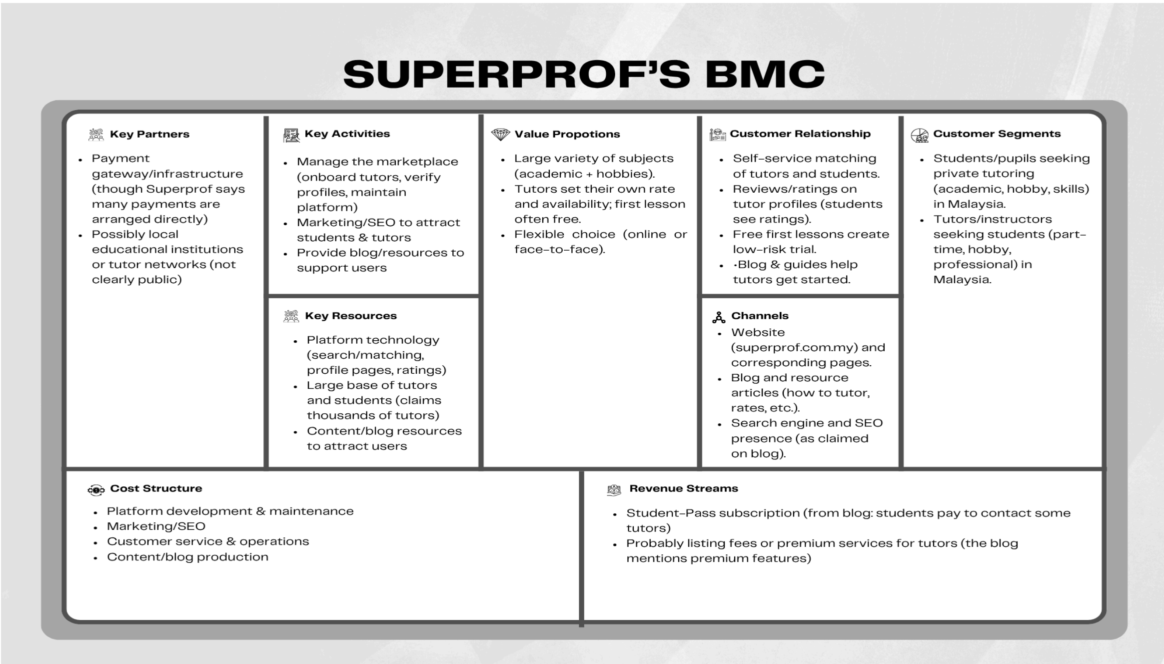


Fig. 1. Business Model Canvas for Superprof

Superprof is an online tutoring platform that connects students with qualified tutors for both academic and non-academic subjects. The platform offers flexibility by allowing learners to choose between online and face-to-face sessions and to filter tutors based on subject, location, price, and availability. Tutors can create their profiles, set their own hourly rates, and often provide a first lesson for free to attract students. Superprof operates on a two-sided marketplace model, serving both students seeking affordable, personalized learning and tutors looking for visibility and teaching opportunities. The platform's value proposition lies in its wide variety of subjects, verified tutor profiles, transparent reviews, and flexible learning formats. In terms of revenue, Superprof adopts a subscription-based model through its Student Pass, which allows students unlimited access to contact tutors for a period. Tutors retain full control of their earnings since Superprof does not take a commission from lesson payments.Overall, Superprof's business model emphasizes accessibility, flexibility, and trust, positioning itself as a user-driven platform that empowers tutors and provides students with convenient access to quality education.

2. GoLearn

GoLearn's BMC

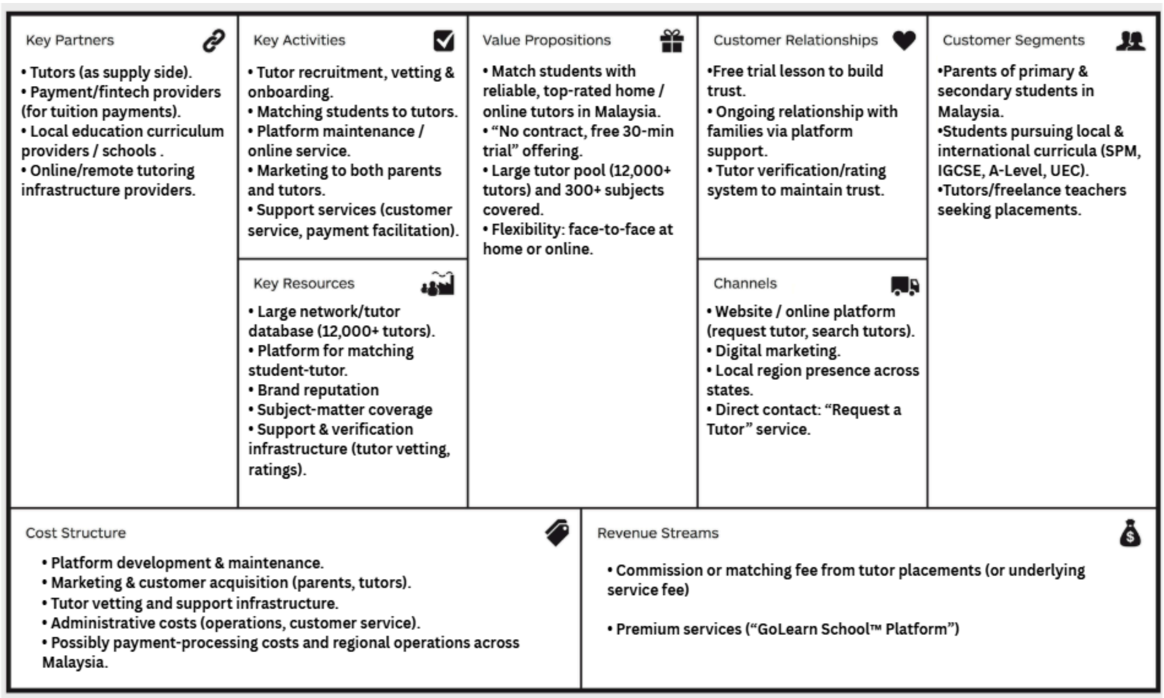


Fig. 2. Business Model Canvas for GoLearn

GoLearn is a Malaysian tutoring-platform brand that connects students/parents with private tutors for home tuition or online tuition lessons across a wide range of subjects and curricula in Malaysia. The platform offers access to a large tutor network covering 300+ subjects and various curricula (SPM, IGCSE, A-Level, UEC, and more), with a key value proposition of flexibility, a free 30-minute trial, and pay-per-lesson convenience. It serves both learners seeking personalised academic support and tutors looking for teaching opportunities, helping ensure a reliable match through screening, ratings, and transparent pricing. With its strong focus on accessibility, convenience, and educational support, GoLearn positions itself as a trusted and scalable solution in Malaysia's growing private education and online learning market.GoLearn's website, while helpful for finding tutors, has limitations in its search and filtering system. It lacks detailed skill-based filters, making it harder for users to identify tutors with specific teaching abilities or specializations. Additionally, the search results are sometimes broad or inaccurate, requiring users to manually sort through many profiles to find a suitable match. These issues can reduce efficiency and user confidence, highlighting the need for more precise search features and clearer tutor skill information.

3. KhanAcademy

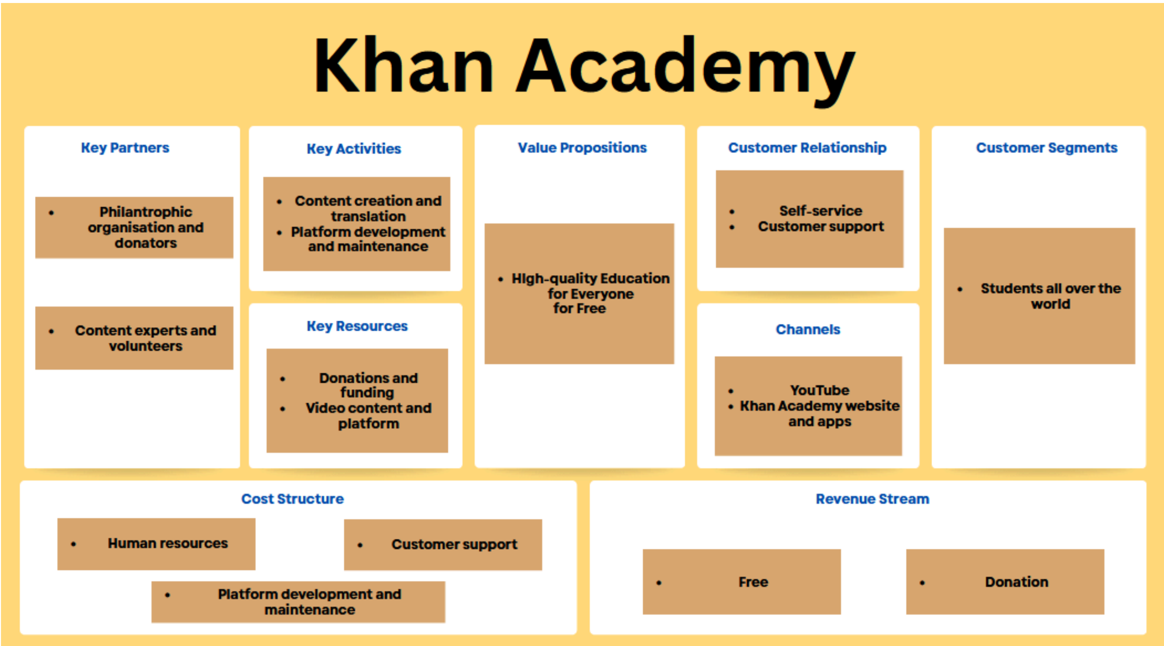


Fig. 3. Business Model Canvas for GoLearn

Khan Academy was founded by Salman "Sal" Khan and officially became a non-profit in 2008. It began in 2004 when Khan, then a hedge fund analyst, started remotely tutoring his cousin. He began posting his tutorial videos on YouTube in 2006, and as their popularity exploded, he quit his job in 2009 to dedicate himself fully to the organization's mission: "to provide a free, world-class education for anyone, anywhere. Khan Academy's business model is primarily donation-based, relying on contributions from philanthropic individuals, foundations, and corporate partners rather than charging users. This approach allows the platform to remain free and accessible to learners worldwide. Additionally, Khan Academy partners with schools, non-profits, and educational organizations to integrate its resources into formal education systems. By offering personalized learning experiences, instant feedback through exercises, and progress tracking, the platform addresses the gaps in traditional classroom learning. While it does not operate for profit, the organization occasionally collaborates with corporate sponsors for specific programs or content development. Overall, its sustainable model focuses on maximizing educational reach and impact, demonstrating that high-quality education can be delivered effectively without financial barriers.

4. Varsity Tutors

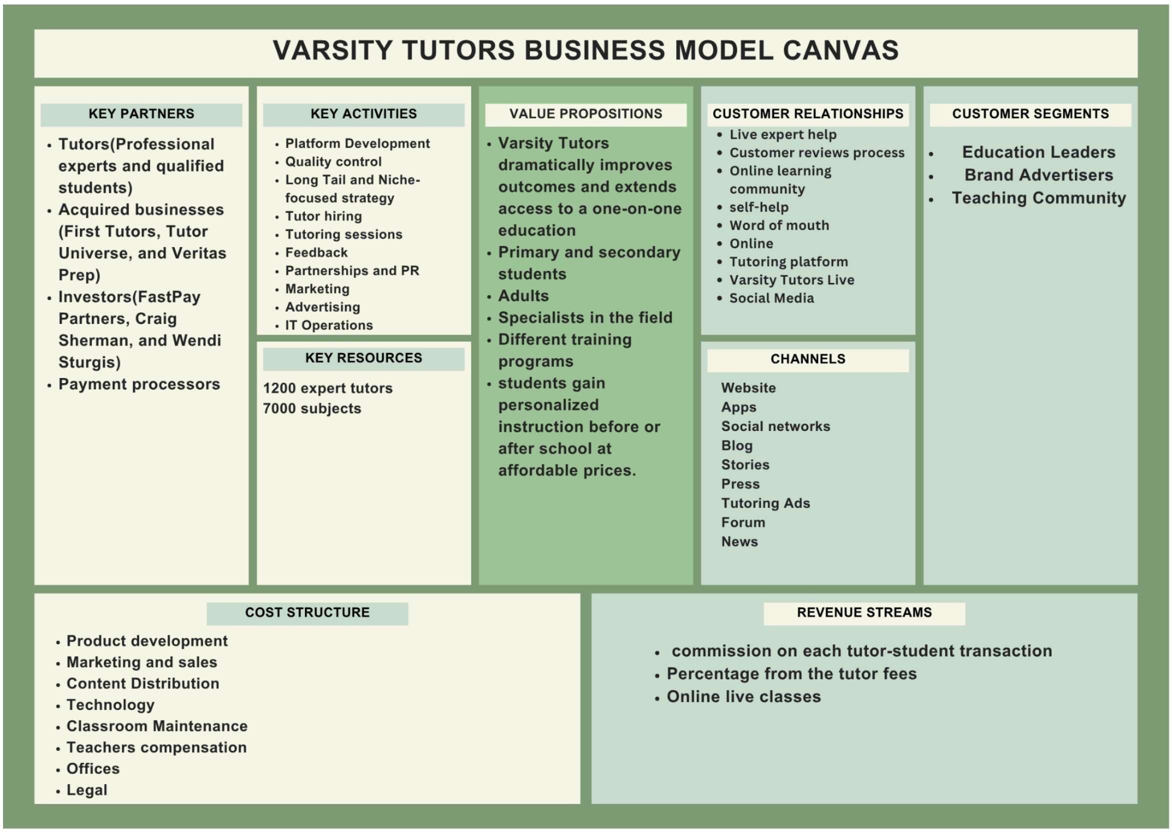


Fig. 4. Business Model Canvas for Varsity Tutors

Varsity Tutors is a comprehensive, live learning platform that connects students and professionals with personalized instruction to achieve any goal. Founded in 2007 and headquartered in St. Louis, Missouri. Its value proposition lies in improving learning outcomes through tailored instruction and extending educational accessibility beyond traditional classroom limitations. There are a few features in Varsity Tutors, including live expert help, integrated feedback systems, and community-based learning spaces.

Although it has been a successful frontrunner in the world of online tutoring, there are some notable issues with Varsity Tutors that make its accessibility and efficiency less than perfect. First, the high service fees and commissions model is priced too high for most users. This may dissuade higher-level, qualified educators from involving themselves in the long run, and also cause less affluent learners to be barred from participating. Moreover, the restricted access to underprivileged users of their platform leads to the problem of educational inequality. Varsity Tutors is quite a bit aimed at students in developed countries, and pricing also subscription plans is usually not affordable for the low-income group. Lastly, Students are not provided with any AI-powered feedback, curated lesson content, or tools to track their progress, and the lack of these features restricts the possibility for personalized learning experiences that continually enhance skills.

5. INITIAL BUSINESS MODEL (BM) – USING BMC & VPC

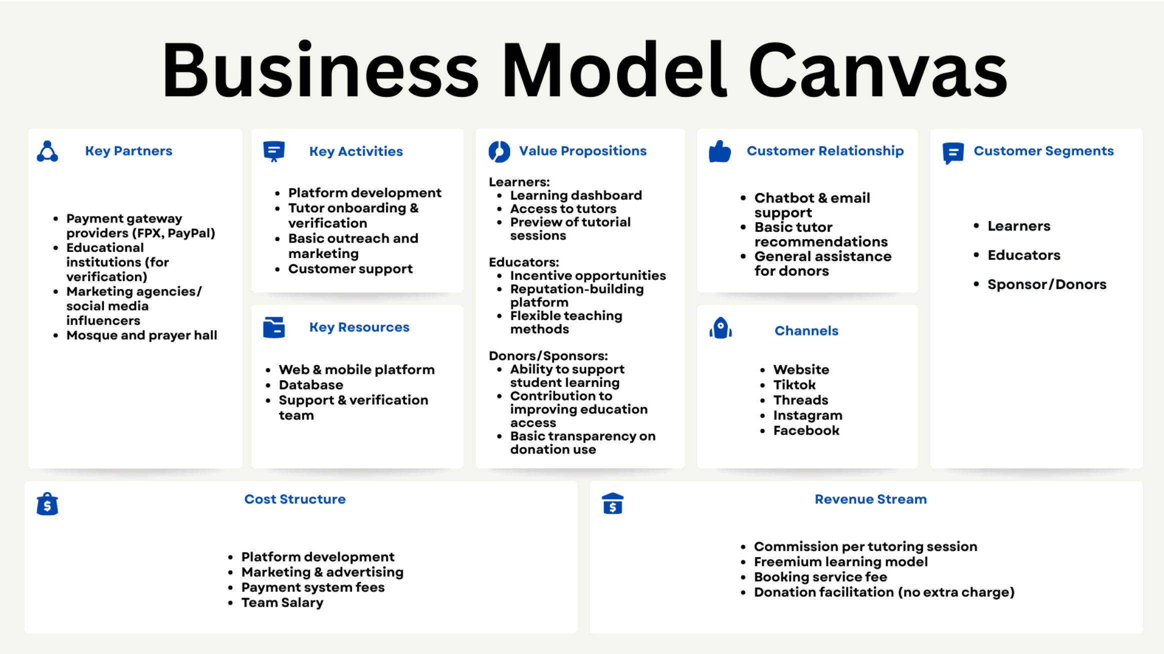


Fig. 5. Initial Business Model Canvas.

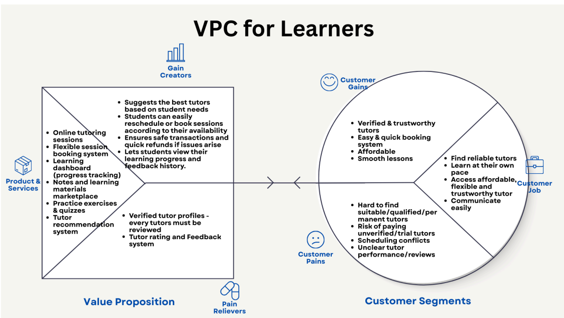


Fig. 6. Value Proposition Canvas for Learners.

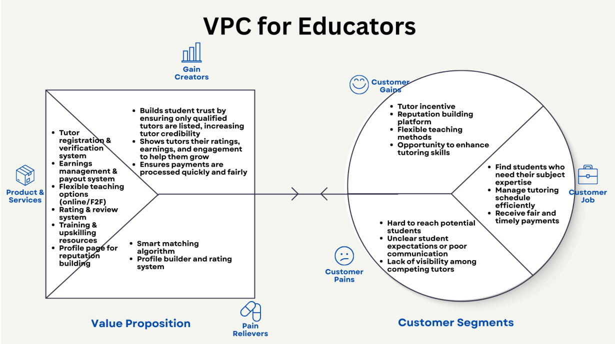


Fig. 7. Value Proposition Canvas for Educators

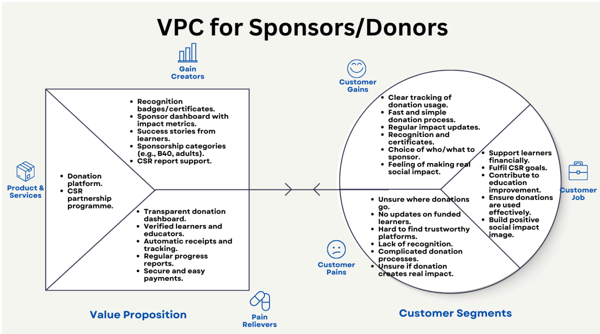


Fig. 7. Value Proposition Canvas for Sponsors/Donors.

6. CONDUCT VALIDATION OF INITIAL BM & KEY FINDINGS

A. Students

Fig. 8. Respondent background

Fig. 9. Student frequency of attending extra classes

Fig. 10. Students' difficulties with tutoring

Fig. 11. Learning methods outside the regular class schedule

Fig. 12. Reasonable price for a one-hour tutoring session according to students

Fig.13. Students' preference to preview tutoring sessions with other students

Fig. 14. Ways trusted tutors are identified

B. Tutor

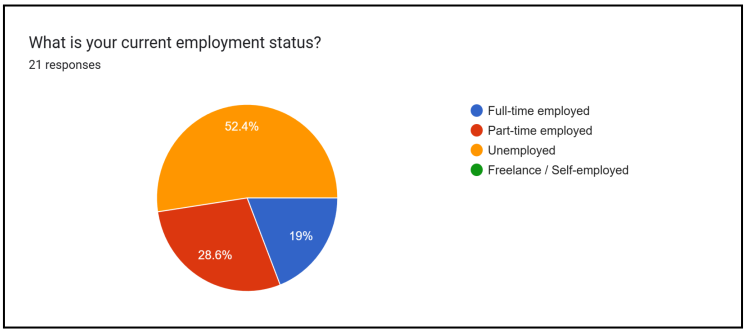


Fig. 15. Percentage of the tutors' employment status.

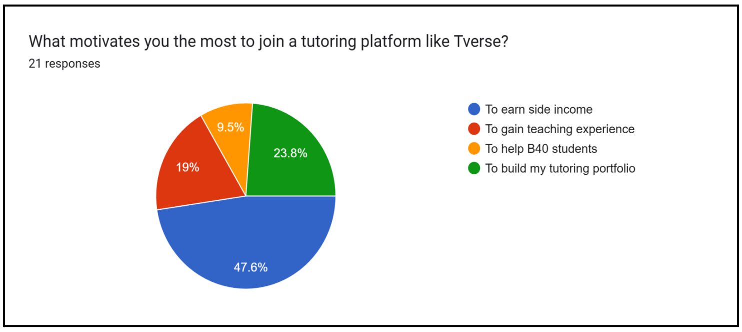


Fig. 16. Percentage of tutors who choose what motivates them to join Tverse.

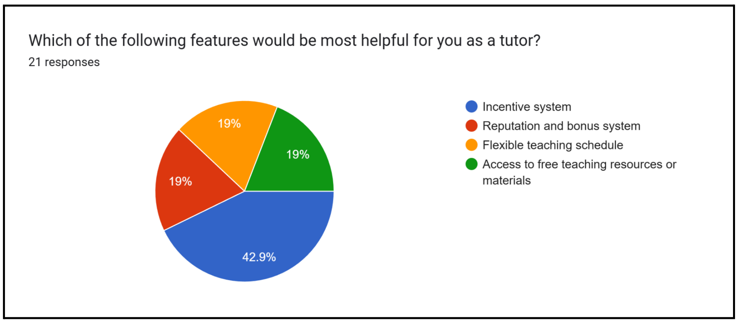


Fig. 17. Percentage of tutors who choose which features are the most helpful.

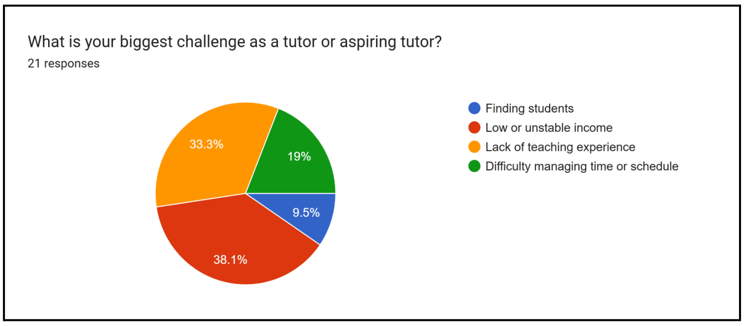


Fig. 18. Percentage of tutors who choose the biggest challenge as a tutor.

Two Google Form surveys have been conducted and distributed to validate our initial business model. The first Google Form survey was designed to understand students' views. From the survey, a majority of respondents were university or college students. Most of them had experience learning through tutoring outside their regular class timetables. Only one respondent did not know where to find a tutor, and only two of the respondents had experienced learning with a private tutor.

Furthermore, respondents preferred lower prices rather than paying based on tutor quality. The respondents were likely to want to see a preview of the tutoring session with other students, with 58.8% voting for this option. There was no clear, significant method for how trusted tutors were by respondents.

Meanwhile, the second Google Form survey was to understand the tutors' points of view. Fig. 15 shows that 21 respondents answered the survey, of whom 11 tutors were unemployed (52.4%), 6 were part-time employed (28.6%), and 4 were full-time employed (19.0%).

A majority of respondents highlighted earning a side income, followed by building a tutoring portfolio, as their main motivations to join Tverse. This validates our decision to include incentive programs and a reputation-building system as core features to help them face their challenges, such as employability and financial issues. Next, many tutors also chose "low or unstable income," followed by "lack of teaching experience," as the main challenges they faced as tutors. This confirms the need for Tverse as their tutoring platform.

6.2 Interviews

Questions for Students :

1. Have you ever used an online tutoring platform before? If yes, how was your overall experience?
2. What do you think are the main benefits of using an online tutoring platform compared to face-to-face learning?
3. What are the biggest problems or challenges you usually face when using online tutoring services?
4. How do you decide whether a tutor is suitable for you on an online platform?
5. What would encourage you to use an online tutoring platform like Tverse more often?

Questions for Tutors :

1. Have you ever taught or offered your tutoring services through an online platform? How was your experience?
2. In your opinion, what are the main advantages of teaching through online tutoring platforms?
3. What challenges do you face when tutoring online (for example, communication, student engagement, payment, or technical issues)?
4. How do you usually attract or connect with new students online?
5. What kind of support or features would make an online tutoring platform more effective and fair for tutors like you?

We interviewed two students, Ahmad Raez and Firaz Hakim, to gather their insights and experiences regarding online tutoring platforms. Raez is a university student who frequently uses online learning tools for academic support, while Firaz is a secondary school student who occasionally seeks online tutoring for exam preparation and subject reinforcement. Both Raez and Firaz shared a generally positive outlook on the use of online tutoring platforms, viewing them as an effective alternative to traditional learning methods due to their flexibility, accessibility, and convenience.

When discussing their challenges, both students mentioned several recurring issues that affect their learning experience. Raez shared that he sometimes struggles with finding reliable and affordable tutors, as some platforms tend to prioritize promotional visibility over tutor quality. He also mentioned that inconsistent internet connections and scheduling conflicts often disrupt his learning flow. Meanwhile, Firaz expressed frustration about limited communication or follow-up from tutors, as well as a lack of local syllabus alignment, especially for Malaysian students preparing for national examinations such as SPM.

Both students agreed that while online tutoring offers great convenience, it can still feel impersonal and difficult to stay motivated without proper engagement strategies from the tutors. Overall, they suggested that Tverse should prioritize verified tutors, consistent lesson quality, affordable pricing models, and an engaging yet simple user interface to ensure a balanced and effective online learning experience for all students.

7. VALIDATED BM – BMC FRAMEWORK

7.1 Business Model Canvas (BMC)

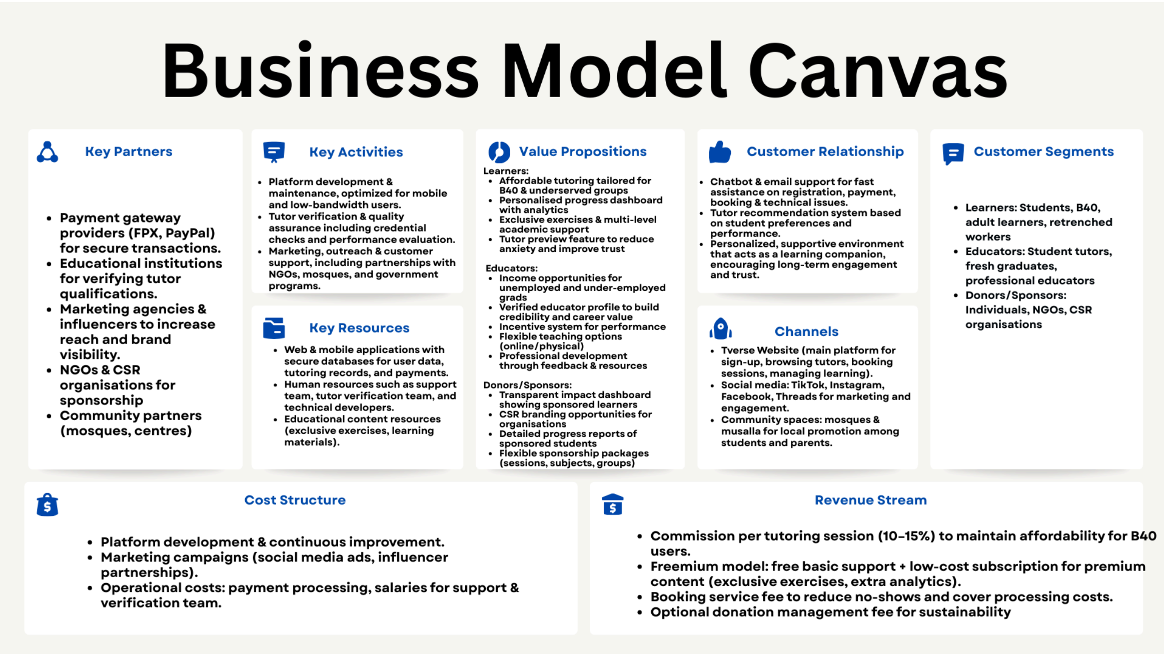


Fig. 5. Enhanced Business Model Canvas.

1. Customer Segments

The Customer Segments (CS) are distinct groups of customers with shared characteristics, such as demographics, behaviour, or needs. In Tverse, the customer segments are (a) learners such as B40 students, adult learners, and retrenched workers who are struggling or need tutoring services from capable tutors and to upskill or reskill themselves, (b) educators from fresh graduates or students who have the experienced to teach and professional educators who are currently facing challenges in securing stable employment and financial issues.

2. Value Proposition

In Business Model Canvas (BMC), Value Proposition (VP) is the core of the business, explaining what unique products or services a company offers to solve a customer's problem or satisfy a need. These are the VPs for Tverse based on each CS:

a. Learners: Tverse is a multi-sided platform that not only serves the normal students but also focuses on B40 students who cannot afford tutoring services, and they need extra support to help them in their education. By using Tverse, each student will have their own dashboard to track their progress, activities, and overall progress. This helps them to identify their strengths, weaknesses, and areas for improvement. Next, the students can preview the tutorial session done by the tutor, which can make them feel eager and motivated to learn using our platform. Tverse also prepared exclusive exercises for the students so that they can practice and study more to achieve their goal. This platform also gives opportunities for the learners, especially adult learners and retrenched workers, to reskill or upskill their tutoring skills, where Tverse provides resources or feedback that help them develop professionally. In Tverse, we also prepared several resources like peer support, expert guidance, or supplementary materials to ensure their comprehensive learning.

b. Educators: The Tverse platform also provides opportunities and side-income jobs for those who have the credibility and a qualified educational background, so that the learners can have a quality tutoring session. Using this platform, the educators can get incentives based on their diligence and performance in tutoring. Not only that, educators can use our platform as a reputation-building tool for their track records, whether to attract more students or to use it for their careers. Educators also will not feel the burden of tutoring using the Tverse platform, as it is a flexible teaching platform where they can choose how to deliver the session, whether in an online session or face-to-face.

c. Sponsors/Donors: Tverse is also open for individuals or organisations who want to make donations or contribute to the community by supporting the underserved learners, especially B40 students. Through Tverse, they can monitor the learner's progress from the dashboard. Organisations can also benefit from Corporate Social Responsibility (CSR) visibility, where their brands are highlighted on the platform. Tverse is also open for flexible sponsorship, which allows donors to contribute based on their capacity and targeted social goals.

3. Channel

Tverse primarily uses digital platform networks to better facilitate the exchange of communications between the learners and tutors, yet remains accessible to users as well as being ubiquitous with one’s brand. Tverse’s website, where users go to sign up, browse tutors, schedule sessions, and manage their learning experience, is the central node. In addition, for broader outreach and engagement, Tverse uses other popular social media platforms, including TikTok, Threads, Instagram, and Facebook, helping it attract the attention of young users and university students in Malaysia. These channels facilitate advertisement, education content sharing, and community involvement in the form of videos, testimonials, and interactive posts from consumers. TVerse would also be promoted through the musalla and mosque, as both are suitable places to spread information. Therefore, Tverse’s strategy ensures both high reach and cost-effective marketing, aligning with the habits of its intended users.

4. Customer Relationship

Tverse is all about creating unique and nurturing connections with its users in order to develop trust with them for the long haul. With user support offered through chatbot and email, the platform promises immediate help for matters concerning registration, payment, or scheduling. Moreover, the system is able to recommend tutors according to students’ profiles, favourite subjects, and performance feedback; it not only boosts user satisfaction but also promotes their retention. This technique was supported by early feedback sessions, where students expressed a strong preference for platforms that feel guided and responsive to individual learning needs. By offering support that is not only helpful but also tailored, along with recommendations, Tverse sets itself apart from being just a tutoring marketplace but rather a learning companion that grows with each user’s progress. This customer-focused structure not only establishes trust but also stimulates repeat usage from both tutors and students.

5. Revenue Streams

Tverse ensures its financial sustainability and social mission through a balanced model that relies on three main Revenue Streams designed to maintain affordability for its B40 segment. The primary income source is a modest Commission per Tutoring Session (a low 10-15% cut), strategically designed to keep effective costs down for students while maintaining high tutor retention. This is supplemented by a Freemium Model that offers basic support for free but captures revenue through low-cost subscriptions for premium learning resources (exclusive exercises and advanced analytics), providing an upgrade path without excluding the core market. Finally, a small Booking Service Fee is charged upon confirmation to minimize scheduling inefficiencies like no-shows and cover transaction processing costs.

6. Key Activities

The successful delivery of Tverse's dual value proposition is predicated on three critical Key Activities. Firstly, Platform Development & Maintenance is essential, requiring continuous technical updates and a specific focus on optimizing for mobile responsiveness and low-bandwidth use to effectively bridge the B40 digital divide. Secondly, Tutor Verification & Quality Control is paramount to guarantee the educational standard; this involves rigorous vetting of academic credentials, background checks, and ongoing performance management driven by real-time student feedback. Lastly, robust Marketing, Outreach & Customer Support efforts are required to drive user acquisition through targeted digital campaigns and establish strategic partnerships with Masjid, NGOs, and government entities to promote the affordable service directly to the target B40 community.

7. Key Resources

TVerse's operations are built on critical technological and human assets. The primary technological resources are its responsive web and mobile application. This will facilitate seamless communication and a secure central database that manages all user data, including tutor profiles, learning progress, and payment histories. On the human side, a dedicated support and verification team is essential for vetting tutors, handling user inquiries, and maintaining the platform's integrity. Together, these resources are foundational to creating a trustworthy, transparent, and effective tutoring environment.

8. Cost Structure

TVerse's primary expenditures are concentrated in three areas: platform development, marketing, and operations. A substantial budget is allocated to continuous technological improvement, including updates and maintenance, to ensure the platform remains secure, intuitive, and competitive. Marketing and user acquisition, particularly campaigns on social media platforms like TikTok, Instagram, and Facebook, represent another major cost center. Finally, ongoing operational expenses include payment processing fees for financial transactions and the costs of running the customer support team. These expenditures are viewed as essential investments to foster long-term user retention and platform growth.

9. Key Partners

To enhance its ecosystem, TVerse engages in strategic collaborations. Key financial partners include payment gateway providers like FPX and PayPal, which ensure all transactions are secure and seamless. Academic partnerships with educational institutions are crucial for validating tutor qualifications, a process that underpins the platform's quality and authenticity. Furthermore, TVerse works with marketing agencies and social media influencers to boost its brand visibility and online presence. These alliances are fundamental to building credibility, broadening the platform's user base, and ensuring sustainable long-term growth.

7.2 Environment Map (EM)

The Environment Map (EM) is a critical strategic tool used to analyze the external landscape that influences a company’s business model design. It provides a structured overview of the major forces that can either enable or constrain the model’s evolution. For Tverse, operating within the fast-moving but socially sensitive Malaysian EdTech sector, understanding these forces is paramount for achieving its dual mandate: delivering educational access and creating economic opportunities for underemployed graduates.

I. Key Trends

This quadrant analyzes the significant societal, technological, and governmental shifts that affect the business model's foundation. Socioeconomic trends, specifically the high incidence of graduate underemployment and the widespread adoption of the gig economy model, ensure a readily available, skilled tutor pool (NHRC, n.d.). Concurrently, the necessity of adapting to the Fourth Industrial Revolution (4IR) and the persistent issue of the digital divide among B40 households demand that Tverse prioritize technical solutions optimized for low-bandwidth and mobile access (Ali & Zainodin, 2024; Devisakti et al., 2023). Regulatory oversight from bodies responsible for ensuring quality in online learning further necessitates rigorous and transparent verification protocols for all tutors (MQA, n.d.).

II. Industry Forces

Industry forces define the competitive structure. Traditional tuition centers (incumbents) offer localized, high-contact services but are constrained by high cost structures, which Tverse effectively eliminates. Generic online platforms (new entrants/substitutes) provide breadth but lack the verified, local, and socio-economic focus that differentiates Tverse. The high supply of potential tutors, driven by economic necessity, mitigates the power of the supplier side, ensuring Tverse maintains a strong position in acquiring Key Resources.

III. Market Forces

Market forces center on customer-level dynamics. Tverse directly addresses the dual demand for affordable, personalized learning (students) and flexible, decent-paying work (tutors). By focusing on affordability and establishing stringent Tutor Verification, Tverse directly confronts the primary market issues of cost barriers and lack of trust, which are typical constraints in low-cost online services.

2.4 IV. Macro-Economic Forces

These forces describe the overarching economic and infrastructure climate. While high mobile penetration across Malaysia supports the digital platform, regional disparities in connectivity and device ownership (economic infrastructure) place design constraints on the platform's user experience. Crucially, Tverse's social mission is to solve a national challenge by making it highly attractive to non-traditional funding avenues, specifically impact investing, which targets measurable socio-economic returns in the education and community sectors (Khazanah Nasional Berhad, n.d.; Securities Commission Malaysia, n.d.).

7.3 Strategy Canvas

The strategy canvas is a strategic visual tool that compares key competing factors across industry players to identify differentiation and competitive advantage (Intrafocus, 2020). In this study, a strategy canvas is developed to position Tverse against similar platforms, namely Superprof, GoLearn Malaysia, Khan Academy, and Varsity Tutors. These platforms represent global and local tutoring and learning competitors across marketplace-based tutoring and free learning content ecosystems.

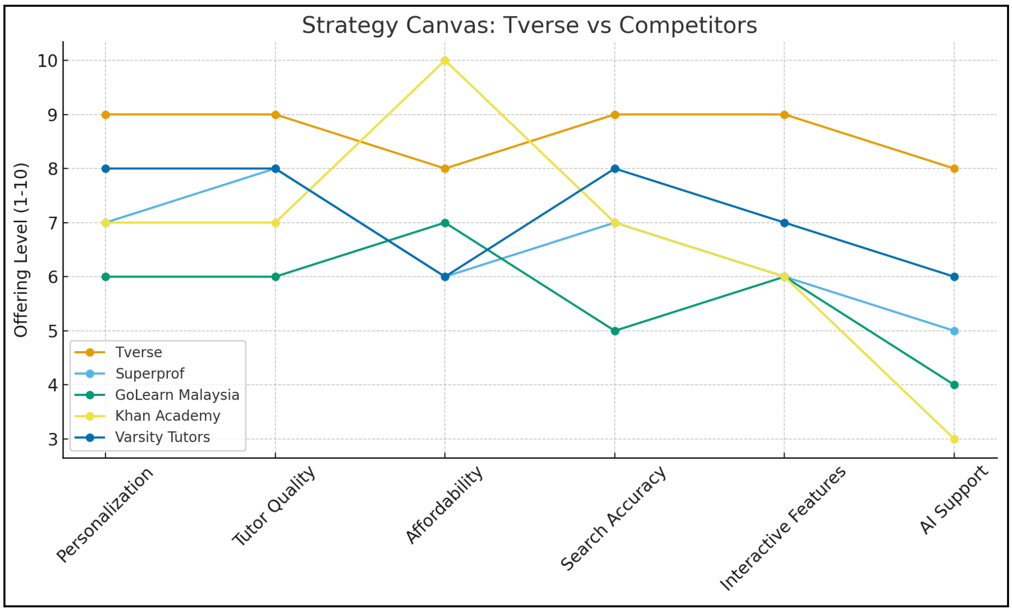
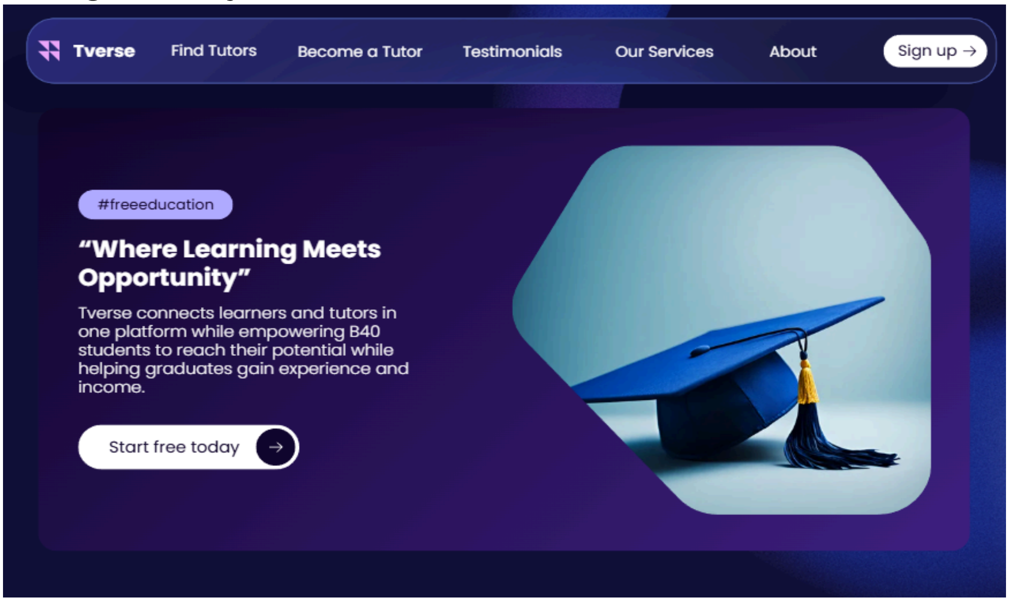
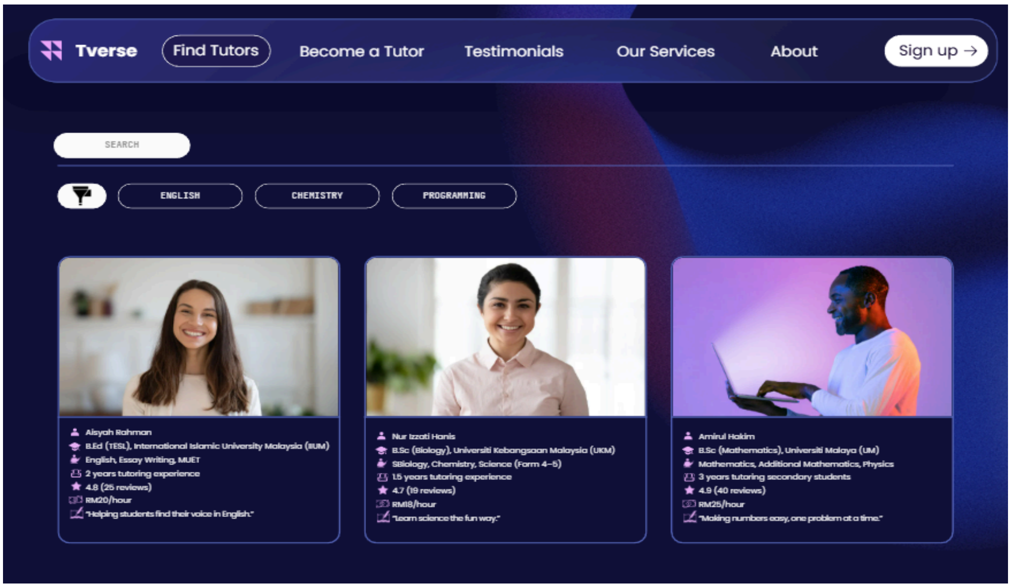


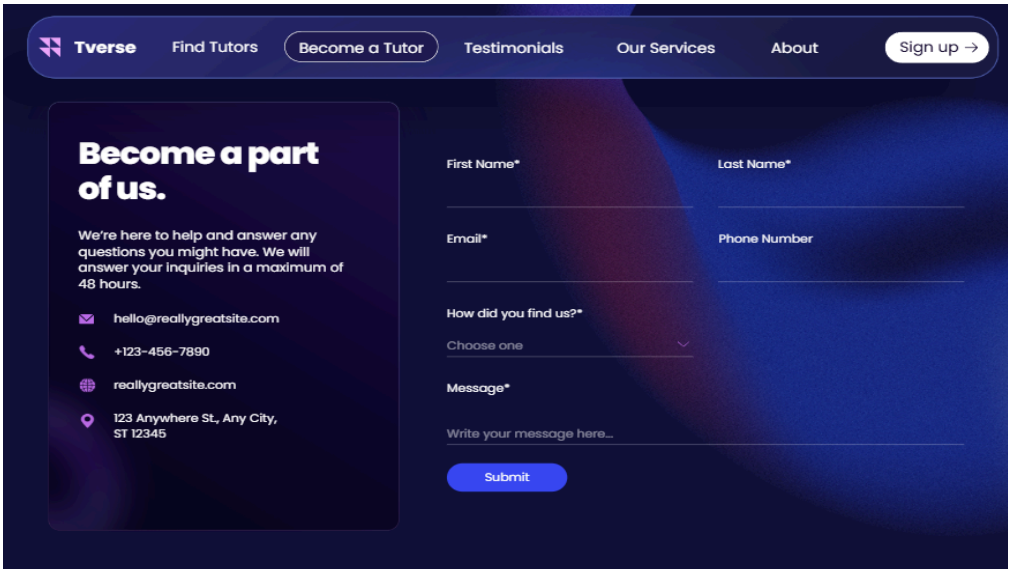
Fig. 19 Strategy Canvas

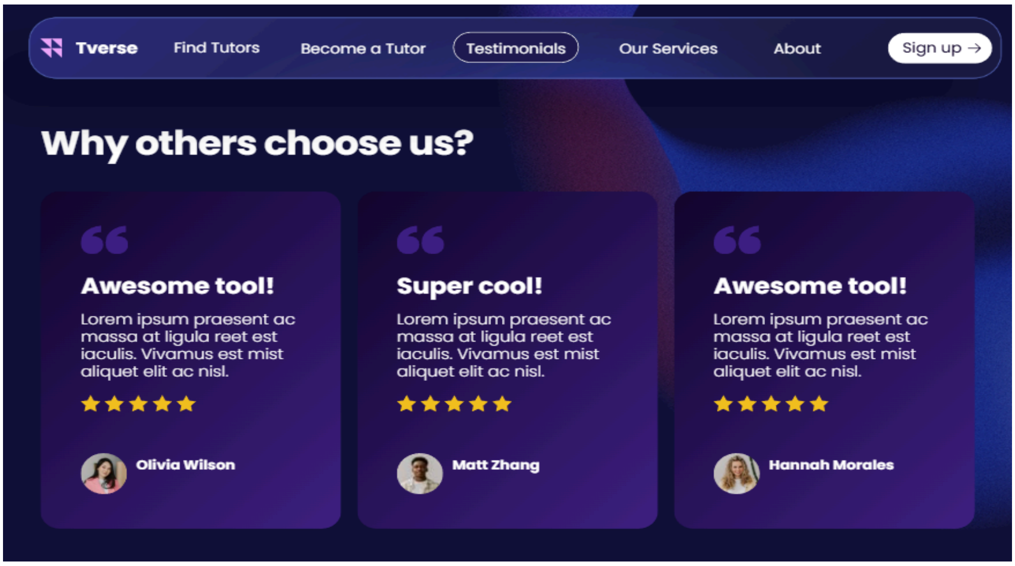
After benchmarking major learning platforms, six key factors were chosen for the Tverse strategy canvas: personalization, tutor quality, affordability, search accuracy, interactive features, and AI support. The strategy canvas illustrates how Tverse positions itself competitively within the digital tutoring landscape by outperforming major learning platforms such as Superprof, GoLearn Malaysia, Khan Academy, and Varsity Tutors across key value dimensions. Tverse demonstrates notably higher offering levels in personalization, tutor quality, search accuracy, and interactive features, reflecting its strong emphasis on tailored learning experiences, verified and capable tutors, progress-tracking dashboards, session previews, and exclusive practice exercises. While Khan Academy leads in affordability due to its free content model, Tverse maintains competitive pricing and uniquely integrates opportunities for unemployed and under-employed graduates to earn income through tutoring, addressing both educational and socioeconomic needs. Although AI support across competitors remains generally moderate, Tverse shows a stronger commitment to leveraging smart recommendations and digital guidance to enhance learning outcomes. Overall, the canvas highlights Tverse's differentiated value proposition as an empowering, student-centered, and socially impactful platform designed to serve B40 learners and aspiring tutors more effectively than traditional tutoring and content-only platforms.

7.4 High Fidelity





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Matt Zhang
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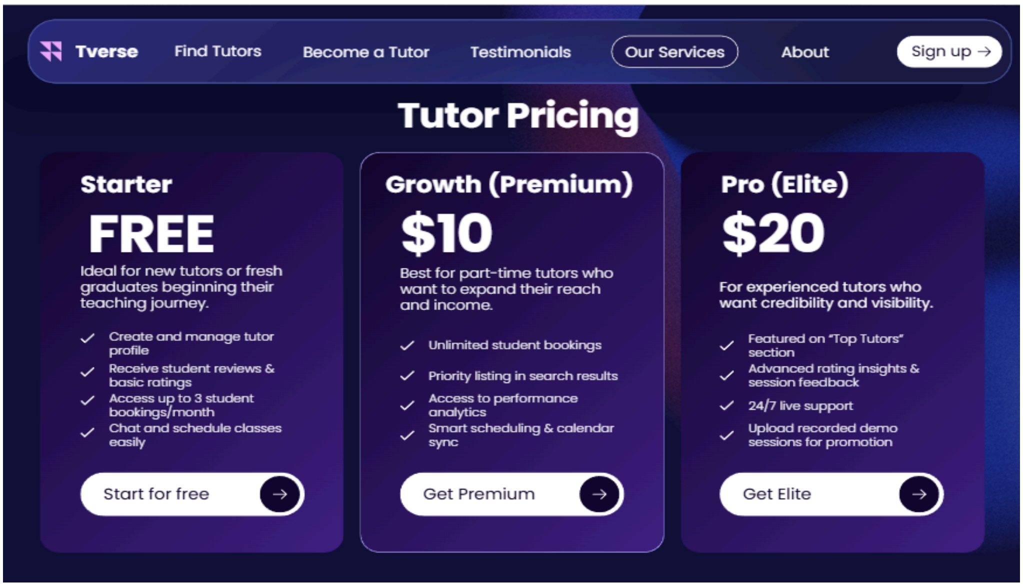
Hannah Morales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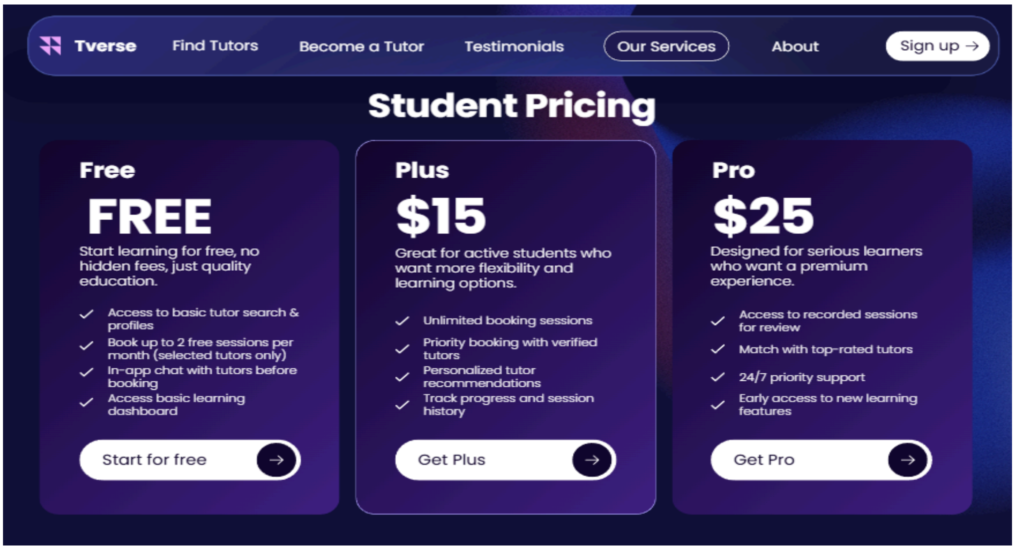
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Book up to 2 free sessions per month (selected tutors only)
In-app chat with tutors before booking
Access basic learning dashboard
Start for free
Plus
$15
Great for active students who want more flexibility and learning options.
Unlimited booking sessions
Priority booking with verified tutors
Personalized tutor recommendations
Track progress and session history
Get Plus
Pro
$25
Designed for serious learners who want a premium experience.
Access to recorded sessions for review
Match with top-rated tutors
24/7 priority support
Early access to new learning features
Get Pro

8. CONCLUSION AND FUTURE WORKS

This study proposed Tverse, a multi-sided digital tutoring platform designed to empower B40 students and unemployed or under-employed graduates by bridging educational gaps and promoting digital entrepreneurship. By applying technopreneurship frameworks, including the Business Model Canvas (BMC), Value Proposition Canvas (VPC), and Strategy Canvas, this paper demonstrated how Tverse can create meaningful value through personalized learning tools, tutor verification, flexible teaching models, and exclusive learning resources. Tverse aligns with SDG 1 (No Poverty) and SDG 4 (Quality Education) by providing affordable academic support and generating income opportunities for skilled graduates, while also fostering social mobility and inclusive access to digital education.

The findings highlight strong market potential for a platform that integrates personalized learning, affordable tutoring, and job creation for graduates, especially in a post-pandemic digital era where online learning and gig-based work continue to grow. Through benchmarking with established platforms such as Superprof, GoLearn Malaysia, Khan Academy, and Varsity Tutors, Tverse demonstrates a differentiated value proposition focused on learning effectiveness, equity, and economic empowerment. As a conceptual model, this work offers a foundation for developing a sustainable learning ecosystem that supports both academic achievement and livelihood enhancement.

Future Works

Moving forward, several steps are recommended to strengthen and operationalize Tverse:

- **Prototype Development**: Build a functional MVP integrating key features such as dashboard tracking, tutor matching, and interactive learning modules.
- **Pilot Testing**: Conduct real-world trials with university students, graduates, and B40 learners to refine user experience and platform usability.
- **AI-Driven Features**: Develop AI recommendation systems for tutor-learner matching, automated performance analytics, and personalized learning pathways.
- **Partnership Expansion**: Collaborate with universities, NGOs, the Ministry of Education, and training agencies to expand credibility and outreach.

- **Funding & Scalability Strategy**: Explore government grants, CSR collaborations, and impact-investment opportunities to scale nationwide.

By pursuing these future directions, Tverse can evolve from a conceptual model into a transformative social-innovation platform, promoting inclusive education, reducing unemployment, and strengthening digital participation for underserved communities.

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